

Applied Bioinformatics Laboratory

Experiment – 4

A: Brain Structure Analysis using BrainMaps

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AIM:

To study and interpret human brain anatomy using the BrainMaps database by examining ultra-high-resolution histological images, with particular emphasis on myelin-stained sagittal sections of the human thalamus.

INTRODUCTION:

BrainMaps.org is an advanced online brain atlas developed for neuroscience research and learning. It offers ultra-high-resolution digital images of brain tissues obtained through histological techniques. The database includes multiple staining methods such as Myelin, Nissl, AChE, CO, and MRI, along with datasets from various species. These features enable detailed visualization of brain regions, fiber tracts, and structural organization. In the present study, the Homo sapiens thalamus dataset stained for myelin was analyzed in sagittal sections to observe myelinated neural pathways and internal thalamic organization.

METHODOLOGY:

1.Access BrainMaps Platform

Open the BrainMaps website at <https://brainmaps.org/>

2.Dataset Selection

- Species: Homo sapiens
- Dataset: Myelin-stained, sagittal histological sections of adult thalamus
- Dataset ID: 98
- Stain type: Myelin
- Section plane: Sagittal
- Brain region: Thalamus

3.Examination of Dataset Information

- Data source: Dr. Edward G. Jones
- Staining technique: Histological myelin staining
- Tissue preparation: Frozen sections
- Image resolution: 0.46 microns per pixel
- Section thickness: 40 microns
- Total number of sections: 16

4.Image Observation

- All available sagittal sections (03a, 03b, 03c, 03d, 05a, 07a, 07b, etc.) were viewed
- Variations in staining intensity were observed based on myelin density

5.Structural Identification

- The thalamus and neighboring brain regions were identified
- Myelinated fib

BrainMaps provides high-resolution brain images for neuroscience research

RESULTS:

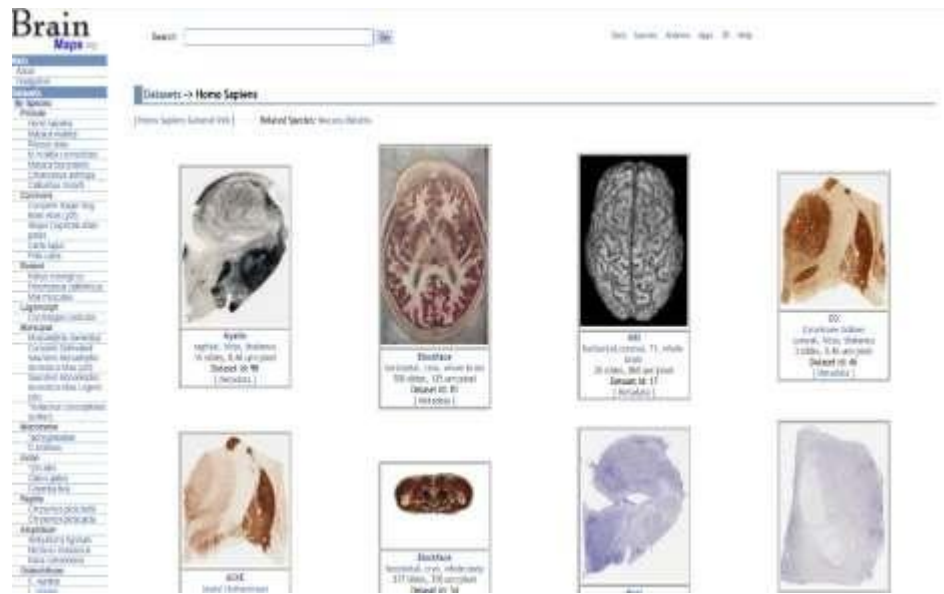


Figure: Human thalamus myelin-stained sagittal sections



Fig 1.2: Sagittal myelin-stained sections of adult human thalamus (BrainMaps)



Fig 1.3: BrainMaps dataset showing sagittal myelin-stained sections of the adult human thalamus

Species	No. of Datasets	Common Stains	Brain Coverage
Human (Homo sapiens)	8	Nissl, Myelin, IHC	Whole brain
Rhesus macaque (Macaca mulatta)	31	Nissl, Myelin	Whole brain
Mouse (Mus musculus)	23	Nissl, Myelin	thalamus
Rat (Rattus norvegicus)	9	Nissl, Myelin	Whole brain
Cat (Felis catus)	6	Nissl, Myelin	Brain stem
Other species	>50	Variable	Whole brain

INTERPRETATION:

The myelin-stained sagittal sections provided a detailed view of the thalamic architecture and surrounding neural pathways. Regions with dense myelination appeared darker, highlighting major fiber tracts and connectivity patterns within the thalamus. The high spatial resolution allowed clear differentiation of structural boundaries and internal organization. These observations demonstrate the effectiveness of BrainMaps as a powerful tool for studying human brain microanatomy and understanding the distribution of myelinated fibers in the thalamus.